

The Problem: How to Choose a Model?

The test set (J_{test}) gives us a good estimate of how well a *single* model generalizes to new data. But what if we are trying to decide *which* model to use in the first place?

For example, we might be choosing between 10 different models:

- Model 1: 1st-degree polynomial ($d = 1$)
 - Model 2: 2nd-degree polynomial ($d = 2$)
 - ...
 - Model 10: 10th-degree polynomial ($d = 10$)
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A Flawed Procedure

A tempting, but flawed, procedure would be:

1. Train all 10 models on the **training set**.
2. Evaluate all 10 models on the **test set**.
3. See which model had the lowest test error (e.g., $d = 5$).
4. Report this lowest test error as our generalization error.

Why this is flawed: We have now used the test set to make a decision—we "fit" the parameter d (the degree of the polynomial) to our test set. By picking the *best* result from this set, our reported test error (J_{test}) is now an **overly optimistic** (artificially low) estimate of how the model will do on new, unseen data.

The Solution: The 3-Way Data Split

To correctly perform model selection, we split our data into **three** separate sets.

1. **Training Set (e.g., 60% of data)**
 - **Purpose:** To train the parameters (w, b) for *each* of our candidate models.
 2. **Cross-Validation (CV) Set (e.g., 20% of data)**
 - **Purpose:** To select the best model. We use this set to "cross-check" the performance of the different models.
 - Also known as the **validation set** or **development (dev) set**.
 3. **Test Set (e.g., 20% of data)**
 - **Purpose:** To provide a final, **unbiased estimate of the generalization error** of our *one* chosen model. This set is "untouched" until the very end.
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The Correct Model Selection Procedure

Here is the standard, correct procedure for choosing the best model and evaluating it fairly:

1. **Train all candidate models** (e.g., all 10 polynomial models) on the **training set**.
2. **Evaluate all trained models** on the **cross-validation (CV) set**.
3. **Pick the one model** that performed the best (had the lowest error) on the **CV set** (J_{cv}). For example, let's say the $d = 4$ polynomial had the lowest CV error.
4. **Finally, take your single chosen model** (the $d = 4$ model) and evaluate it *one time* on the **test set**.
5. Report the resulting J_{test} as the unbiased estimate of your model's generalization error.

This same procedure is used for choosing any model parameters, such as the architecture of a neural network (e.g., comparing a small, medium, and large network to see which performs best on the CV set).

Best Practice: All decisions—fitting parameters (w, b), choosing the model (polynomial degree d , network architecture, regularization λ)—should be made using *only* the **training set** and **cross-validation set**. The **test set** should only be used once at the very end to get a final performance metric.